D. Srinivas Rao, 24EG104A06, ECE-A

Sky130HD • Post-route timing • 16 PVT corners • 12 ns clock (83.33 MHz)

1. Objective and Library Interpretation

Static Timing Analysis (STA) evaluates whether signals reach their destination registers within the clock period under different Process, Voltage and Temperature (PVT) conditions. Each Sky130HD Liberty (.lib) file provides cell delays, setup/hold constraints and electrical characteristics for one operating corner.

Example: sky130_fd_sc_hd__ss_n40C_1v28.lib = Sky130 high-density cells, slow-slow process, −40°C and 1.28 V. FF = fast-fast; SS = slow-slow; TT = typical-typical. Higher voltage generally reduces delay, while temperature effects can vary.

2. Reading a Timing Report

Startpoint = launching flip-flop; Endpoint = capturing flip-flop; Path Type max = setup analysis. Delay is each incremental cell/arc delay; Time is cumulative arrival time; ^ and v indicate rising and falling transitions. Required time is the capture deadline after setup time. Slack = required time − arrival time: positive = MET, negative = VIOLATED.

3. Corner-by-Corner Results

Insert the matching OpenROAD report screenshot in each marked space. The order below matches the original terminal output. All values are for the single path shown by report_checks at that corner.

run.tcl file used:

```
# ============================================================
# VSDBabySoC - Multi-Corner STA
# Terminal Output Only
# ============================================================
set base [pwd]
set lib_dir "$base/platforms/sky130hd/lib"
set lef_dir "$base/platforms/sky130hd/lef"
set macro_lib_dir "$base/designs/sky130hd/vsdbabysoc/lib"
set macro_lef_dir "$base/designs/sky130hd/vsdbabysoc/lef"
# ============================================================
# All 16 PVT Libraries
# ============================================================
set libs {
  sky130_fd_sc_hd__ff_100C_1v65.lib
  sky130_fd_sc_hd__ff_100C_1v95.lib
  sky130_fd_sc_hd__ff_n40C_1v56.lib
  sky130_fd_sc_hd__ff_n40C_1v65.lib
  sky130_fd_sc_hd__ff_n40C_1v76.lib
  sky130_fd_sc_hd__ff_n40C_1v95.lib
  sky130_fd_sc_hd__ss_100C_1v40.lib
  sky130_fd_sc_hd__ss_100C_1v60.lib
  sky130_fd_sc_hd__ss_n40C_1v28.lib
  sky130_fd_sc_hd__ss_n40C_1v35.lib
  sky130_fd_sc_hd__ss_n40C_1v40.lib
  sky130_fd_sc_hd__ss_n40C_1v44.lib
  sky130_fd_sc_hd__ss_n40C_1v60.lib
  sky130_fd_sc_hd__ss_n40C_1v76.lib
  sky130_fd_sc_hd__tt_025C_1v80.lib
  sky130_fd_sc_hd__tt_100C_1v80.lib
}
# ============================================================
# Define Timing Corners
# ============================================================
set corners {}
foreach lib $libs {
  lappend corners [file rootname $lib]
}
define_corners {*}$corners
# ============================================================
# Load LEF Files
# ============================================================
read_lef "$lef_dir/sky130_fd_sc_hd.tlef"
read_lef "$lef_dir/sky130_fd_sc_hd_merged.lef"
foreach macro {avsddac avsdpll} {
  set lef_file "$macro_lef_dir/${macro}.lef"
  if {[file exists $lef_file]} {
    read_lef $lef_file
  }
}
```

```
# ============================================================
# Load Liberty Libraries
# ============================================================
foreach lib $libs {
  set corner [file rootname $lib]
  puts "Loading corner: $corner"
  read_liberty -corner $corner "$lib_dir/$lib"
  foreach macro {avsddac avsdpll} {

    set macro_file "$macro_lib_dir/${macro}.lib"

    set f [open $macro_file r]
    set content [read $f]
    close $f

    regsub -all \
      {library[ \t]*\([ \t]*([a-zA-Z0-9_]+)[ \t]*\)} \
      $content \
      "library (${macro}_${corner})" \
      content

    set tmp_file "/tmp/${macro}_${corner}.lib"

    set f [open $tmp_file w]
    puts -nonewline $f $content
    close $f

    read_liberty -corner $corner $tmp_file

    file delete $tmp_file
  }
}
# ============================================================
# Load Synthesized Netlist
# ============================================================
read_verilog "$base/results/sky130hd/vsdbabysoc/base/1_synth.v"
link_design vsdbabysoc
# ============================================================
# Load Timing Constraints
# ============================================================
read_sdc "$base/designs/sky130hd/vsdbabysoc/vsdbabysoc_synthesis.sdc"
# ============================================================
# Display STA Reports in Terminal
# ============================================================
foreach lib $libs {
  set corner [file rootname $lib]
  puts "\n========================================="
  puts "STA CORNER: $corner"
  puts "========================================="
```

```
  puts "\n========== REPORT CHECKS =========="
  report_checks -corner $corner

  puts "\n========== WNS =========="
  report_worst_slack -max

  puts "\n========== TNS =========="
  report_tns -max
}

puts "\n========================================="
puts "STA COMPLETED FOR ALL 16 PVT CORNERS"
puts "========================================="
```

Working of the run.tcl File

The run.tcl file automates Static Timing Analysis of VSDBabySoC across **16 Sky130HD PVT corners** using OpenROAD.

1. **Path Initialization:** Defines directories containing Liberty (.lib), LEF, synthesized netlist, and SDC files.
2. **Corner Creation:** Uses a foreach loop to create and configure all 16 PVT corners, covering FF, SS, and TT conditions.
3. **Library Loading:** Loads the corresponding standard-cell Liberty file for each corner. DAC and PLL Liberty library names are modified to avoid duplicate-name conflicts.
4. **Design Initialization:** Loads the LEF files, imports 1_synth.v, and links the vsdbabysoc top module.
5. **Timing Constraints:** Reads vsdbabysoc_synthesis.sdc, which defines the 12 ns clock period and other timing constraints.
6. **STA Execution:** Executes report_checks for each corner to display propagation delays, arrival times, required times, and setup slack. WNS and TNS commands report the overall timing violations.

Execution Command:

openroad -exit designs/sky130hd/vsdbabysoc/run.tcl

The script displays all 16 corner results sequentially in the terminal, eliminating the need to run each analysis manually.

1. FF | 100°C | 1.65 V

Arrival: 2.64 ns Slack: 9.26 ns Status: MET. At 1.65 V the path meets setup timing with a large margin.

```
Corner: sky130_fd_sc_hd__ff_100C_1v65

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.35    0.35 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.42    0.77 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.17    0.95 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.22    1.17 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.41    1.58 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.13    1.71 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.06    1.77 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.18    1.95 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.08    2.03 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.10    2.12 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.06    2.19 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.23    2.41 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.17    2.59 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.05    2.64 v _08777_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    2.64 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.64   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   11.90   library setup time
          11.90   data required time
---------------------------------------------------------
          11.90   data required time
          -2.64   data arrival time
---------------------------------------------------------
           9.26   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

2. FF | 100°C | 1.95 V

Arrival: 2.05 ns Slack: 9.88 ns Status: MET. Higher voltage reduces arrival time and improves margin.

```
Corner: sky130_fd_sc_hd__ff_100C_1v95

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.34    0.62 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.13    0.75 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.18    0.93 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.30    1.23 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.10    1.33 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.05    1.38 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.14    1.52 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.05    1.57 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.08    1.65 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.05    1.70 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.18    1.88 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.13    2.01 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.04    2.05 v _09069_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    2.05 v core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.05   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[1][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.07   11.93   library setup time
          11.93   data required time
---------------------------------------------------------
          11.93   data required time
          -2.05   data arrival time
---------------------------------------------------------
           9.88   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

3. FF | –40°C | 1.56 V

Arrival: 3.46 ns Slack: 8.40 ns Status: MET. Lowest reported FF margin, but timing still passes.

```
Corner: sky130_fd_sc_hd__ff_n40C_1v56

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.42    0.42 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.58    1.01 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.24    1.24 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.27    1.52 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.55    2.07 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.18    2.25 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.07    2.32 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.25    2.57 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.09    2.66 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.13    2.80 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.07    2.86 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.32    3.18 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.23    3.41 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.05    3.46 v _08777_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    3.46 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.46   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.15   11.85   library setup time
          11.85   data required time
---------------------------------------------------------
          11.85   data required time
          -3.46   data arrival time
---------------------------------------------------------
           8.40   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

4. FF | –40°C | 1.65 V

Arrival: 2.98 ns Slack: 8.90 ns Status: MET. Increased voltage improves propagation delay.

```
Corner: sky130_fd_sc_hd__ff_n40C_1v65

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.38    0.38 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.51    0.88 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.19    1.08 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.24    1.32 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.46    1.78 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.16    1.94 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.06    2.00 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.22    2.22 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.07    2.29 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.12    2.41 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.06    2.47 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.28    2.74 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.19    2.93 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.04    2.98 v _08777_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    2.98 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.98   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   11.88   library setup time
          11.88   data required time
---------------------------------------------------------
          11.88   data required time
          -2.98   data arrival time
---------------------------------------------------------
           8.90   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

5. FF | –40°C | 1.76 V

Arrival: 2.56 ns Slack: 9.34 ns Status: MET. Timing margin increases further at 1.76 V.

```
Corner: sky130_fd_sc_hd__ff_n40C_1v76

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.33    0.33 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.44    0.77 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.16    0.93 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.22    1.15 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.38    1.53 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.14    1.66 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.05    1.71 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.19    1.90 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.06    1.96 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.10    2.06 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.05    2.11 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.24    2.36 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.16    2.52 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.04    2.56 v _08777_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    2.56 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.56   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.10   11.90   library setup time
          11.90   data required time
---------------------------------------------------------
          11.90   data required time
          -2.56   data arrival time
---------------------------------------------------------
           9.34   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

6. FF | –40°C | 1.95 V

Arrival: 2.08 ns Slack: 9.84 ns Status: MET. Fast propagation and comfortable setup margin.

```
Corner: sky130_fd_sc_hd__ff_n40C_1v95

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.28    0.28 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.36    0.65 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.12    0.77 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.18    0.95 ^ _07849_/X (sky130_fd_sc_hd__buf_2)
   0.29    1.25 v _07853_/X (sky130_fd_sc_hd__mux4_2)
   0.11    1.36 ^ _07854_/Y (sky130_fd_sc_hd__mux2i_1)
   0.04    1.40 v _08245_/Y (sky130_fd_sc_hd__o32ai_1)
   0.15    1.55 ^ _08253_/Y (sky130_fd_sc_hd__nor3_1)
   0.05    1.60 v _08254_/Y (sky130_fd_sc_hd__nand3_1)
   0.08    1.68 ^ _08268_/Y (sky130_fd_sc_hd__a31oi_1)
   0.04    1.72 v _08269_/Y (sky130_fd_sc_hd__a21oi_1)
   0.20    1.92 ^ _08270_/Y (sky130_fd_sc_hd__mux2i_1)
   0.13    2.05 ^ _08775_/X (sky130_fd_sc_hd__buf_2)
   0.03    2.08 v _08777_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    2.08 v core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           2.08   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][24]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.08   11.92   library setup time
          11.92   data required time
---------------------------------------------------------
          11.92   data required time
          -2.08   data arrival time
---------------------------------------------------------
           9.84   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

7. SS | 100°C | 1.40 V

Arrival: 9.93 ns Slack: 1.57 ns Status: MET. Low voltage and slow process leave little timing margin.

```
Corner: sky130_fd_sc_hd__ss_100C_1v40

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.18    1.18 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.24    2.42 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.72    3.14 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.42    3.56 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   1.77    5.33 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   1.60    6.94 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.57    7.51 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.55    8.06 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.38    8.44 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.47    8.90 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.45    9.35 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.44    9.79 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.14    9.93 v _09081_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00    9.93 v core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           9.93   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.50   11.50   library setup time
          11.50   data required time
---------------------------------------------------------
          11.50   data required time
          -9.93   data arrival time
---------------------------------------------------------
           1.57   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

8. SS | 100°C | 1.60 V

Arrival: 6.76 ns Slack: 4.95 ns Status: MET. Higher voltage substantially improves the margin.

```
Corner: sky130_fd_sc_hd__ss_100C_1v60

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.83    0.83 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.88    1.71 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.48    2.19 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.30    2.49 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   1.15    3.65 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   1.02    4.67 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.38    5.05 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.39    5.43 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.26    5.69 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.33    6.02 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.32    6.35 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.31    6.66 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.10    6.76 v _08894_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00    6.76 v core.CPU_Xreg_value_a4[16][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           6.76   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[16][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.30   11.70   library setup time
          11.70   data required time
---------------------------------------------------------
          11.70   data required time
          -6.76   data arrival time
---------------------------------------------------------
           4.95   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

9. SS | −40°C | 1.28 V

Arrival: 27.48 ns Slack: −17.16 ns Status: VIOLATED. Worst reported path: data arrives 17.16 ns after its deadline.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v28

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.82    2.82 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   3.43    6.25 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   2.07    8.32 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   1.03    9.35 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   5.33   14.68 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   4.94   19.61 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   1.46   21.08 v _08467_/X (sky130_fd_sc_hd__and3_1)
   1.68   22.75 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.82   23.58 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   1.09   24.67 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   1.38   26.05 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   1.16   27.21 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.27   27.48 v _08792_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00   27.48 v core.CPU_Xreg_value_a4[11][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          27.48   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.68   10.32   library setup time
          10.32   data required time
---------------------------------------------------------
          10.32   data required time
         -27.48   data arrival time
---------------------------------------------------------
         -17.16   slack (VIOLATED)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

10. SS | –40°C | 1.35 V

Arrival: 18.81 ns Slack: –7.89 ns Status: VIOLATED. Violation decreases as voltage rises, but persists.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v35

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   2.01    2.01 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.48    4.48 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   1.44    5.92 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.74    6.66 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   3.40   10.06 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   3.27   13.33 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   1.00   14.33 v _08467_/X (sky130_fd_sc_hd__and3_1)
   1.16   15.49 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.58   16.07 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.76   16.84 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.95   17.79 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.81   18.61 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.20   18.81 v _08894_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00   18.81 v core.CPU_Xreg_value_a4[16][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          18.81   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[16][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -1.08   10.92   library setup time
          10.92   data required time
---------------------------------------------------------
          10.92   data required time
         -18.81   data arrival time
---------------------------------------------------------
          -7.89   slack (VIOLATED)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

11. SS | –40°C | 1.40 V

Arrival: 15.11 ns Slack: –3.95 ns Status: VIOLATED. The path remains too slow for the 12 ns clock.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v40

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.64    1.64 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   2.02    3.66 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   1.15    4.81 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.62    5.43 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   2.70    8.13 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   2.56   10.69 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.80   11.49 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.94   12.43 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.47   12.90 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.63   13.52 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.77   14.29 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.66   14.95 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.16   15.11 v _08792_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00   15.11 v core.CPU_Xreg_value_a4[11][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          15.11   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.84   11.16   library setup time
          11.16   data required time
---------------------------------------------------------
          11.16   data required time
         -15.11   data arrival time
---------------------------------------------------------
          -3.95   slack (VIOLATED)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

12. SS | –40°C | 1.44 V

Arrival: 12.95 ns Slack: −1.65 ns Status: VIOLATED. Nearly meets timing, but still misses the deadline.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v44

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   1.43    1.43 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.74    3.17 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.98    4.15 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.53    4.68 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   2.29    6.97 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   2.16    9.12 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.69    9.81 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.81   10.62 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.41   11.03 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.55   11.58 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.66   12.24 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.57   12.81 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.14   12.95 v _09081_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00   12.95 v core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
          12.95   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.70   11.30   library setup time
          11.30   data required time
---------------------------------------------------------
          11.30   data required time
         -12.95   data arrival time
---------------------------------------------------------
          -1.65   slack (VIOLATED)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

13. SS | –40°C | 1.60 V

Arrival: 8.08 ns Slack: 3.53 ns Status: MET. The path returns to positive setup slack.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v60

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.93    0.93 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   1.14    2.07 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.60    2.66 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.36    3.02 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   1.37    4.39 v _08102_/X (sky130_fd_sc_hd__mux4_2)
   1.25    5.64 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.42    6.07 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.51    6.58 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.25    6.84 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.36    7.20 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.43    7.63 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.36    7.99 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.09    8.08 v _09176_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00    8.08 v core.CPU_Xreg_value_a4[26][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           8.08   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[26][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.39   11.61   library setup time
          11.61   data required time
---------------------------------------------------------
          11.61   data required time
          -8.08   data arrival time
---------------------------------------------------------
           3.53   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

14. SS | –40°C | 1.76 V

Arrival: 5.78 ns Slack: 5.97 ns Status: MET. Best reported SS margin at −40°C.

```
Corner: sky130_fd_sc_hd__ss_n40C_1v76

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_is_slli_a3$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.69    0.69 ^ core.CPU_is_slli_a3$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.83    1.52 ^ _07545_/Y (sky130_fd_sc_hd__mux2i_4)
   0.42    1.94 ^ _07555_/X (sky130_fd_sc_hd__buf_6)
   0.26    2.20 ^ _07671_/X (sky130_fd_sc_hd__buf_6)
   0.93    3.13 v _08275_/X (sky130_fd_sc_hd__mux4_2)
   0.86    3.99 v _08466_/X (sky130_fd_sc_hd__mux4_2)
   0.30    4.29 v _08467_/X (sky130_fd_sc_hd__and3_1)
   0.37    4.67 ^ _08469_/Y (sky130_fd_sc_hd__a311oi_1)
   0.19    4.85 v _08470_/Y (sky130_fd_sc_hd__o211ai_1)
   0.27    5.12 v _08483_/X (sky130_fd_sc_hd__o211a_2)
   0.32    5.44 ^ _08484_/Y (sky130_fd_sc_hd__mux2i_4)
   0.27    5.71 ^ _08790_/X (sky130_fd_sc_hd__buf_6)
   0.07    5.78 v _09081_/Y (sky130_fd_sc_hd__a21oi_2)
   0.00    5.78 v core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           5.78   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[1][29]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.25   11.75   library setup time
          11.75   data required time
---------------------------------------------------------
          11.75   data required time
          -5.78   data arrival time
---------------------------------------------------------
           5.97   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

15. TT | 25°C | 1.80 V

Arrival: 3.41 ns Slack: 8.46 ns Status: MET. Nominal room-temperature corner meets timing.

```
Corner: sky130_fd_sc_hd__tt_025C_1v80

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.33    0.33 ^ core.CPU_src2_value_a3[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.07    0.40 v _05540_/Y (sky130_fd_sc_hd__inv_1)
   0.26    0.67 ^ _10845_/SUM (sky130_fd_sc_hd__ha_1)
   0.28    0.95 ^ _05493_/X (sky130_fd_sc_hd__clkbuf_1)
   0.10    1.05 v _07516_/Y (sky130_fd_sc_hd__nor2b_1)
   0.39    1.44 ^ _07518_/Y (sky130_fd_sc_hd__a211oi_1)
   0.12    1.55 v _07521_/Y (sky130_fd_sc_hd__nand3_1)
   0.27    1.83 ^ _07524_/Y (sky130_fd_sc_hd__a41oi_1)
   0.16    1.99 v _08159_/Y (sky130_fd_sc_hd__o21ai_0)
   0.36    2.35 v _08323_/X (sky130_fd_sc_hd__a221o_1)
   0.26    2.61 v _08330_/X (sky130_fd_sc_hd__o221a_2)
   0.22    2.83 v _08348_/X (sky130_fd_sc_hd__o22a_1)
   0.30    3.13 ^ _08349_/Y (sky130_fd_sc_hd__mux2i_1)
   0.21    3.35 ^ _08781_/X (sky130_fd_sc_hd__buf_2)
   0.06    3.41 v _08783_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    3.41 v core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.41   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.13   11.87   library setup time
          11.87   data required time
---------------------------------------------------------
          11.87   data required time
          -3.41   data arrival time
---------------------------------------------------------
           8.46   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

16. TT | 100°C | 1.80 V

Arrival: 3.32 ns Slack: 8.56 ns Status: MET. Slightly shorter delay than TT at 25°C; timing passes.

```
Corner: sky130_fd_sc_hd__tt_100C_1v80

  Delay    Time   Description
---------------------------------------------------------
   0.00    0.00   clock clk (rise edge)
   0.00    0.00   clock network delay (ideal)
   0.00    0.00 ^ core.CPU_src2_value_a3[4]$_DFF_P_/CLK (sky130_fd_sc_hd__dfxtp_1)
   0.33    0.33 ^ core.CPU_src2_value_a3[4]$_DFF_P_/Q (sky130_fd_sc_hd__dfxtp_1)
   0.08    0.41 v _05540_/Y (sky130_fd_sc_hd__inv_1)
   0.26    0.67 ^ _10845_/SUM (sky130_fd_sc_hd__ha_1)
   0.28    0.95 ^ _05493_/X (sky130_fd_sc_hd__clkbuf_1)
   0.10    1.06 v _07516_/Y (sky130_fd_sc_hd__nor2b_1)
   0.36    1.42 ^ _07518_/Y (sky130_fd_sc_hd__a211oi_1)
   0.12    1.54 v _07521_/Y (sky130_fd_sc_hd__nand3_1)
   0.26    1.80 ^ _07524_/Y (sky130_fd_sc_hd__a41oi_1)
   0.17    1.97 v _08159_/Y (sky130_fd_sc_hd__o21ai_0)
   0.34    2.31 v _08323_/X (sky130_fd_sc_hd__a221o_1)
   0.24    2.55 v _08330_/X (sky130_fd_sc_hd__o221a_2)
   0.22    2.77 v _08348_/X (sky130_fd_sc_hd__o22a_1)
   0.28    3.05 ^ _08349_/Y (sky130_fd_sc_hd__mux2i_1)
   0.21    3.26 ^ _08781_/X (sky130_fd_sc_hd__buf_2)
   0.06    3.32 v _08783_/Y (sky130_fd_sc_hd__a21oi_1)
   0.00    3.32 v core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/D (sky130_fd_sc_hd__dfxtp_1)
           3.32   data arrival time

  12.00   12.00   clock clk (rise edge)
   0.00   12.00   clock network delay (ideal)
   0.00   12.00   clock reconvergence pessimism
          12.00 ^ core.CPU_Xreg_value_a4[11][26]$_SDFFE_PP0P_/CLK (sky130_fd_sc_hd__dfxtp_1)
  -0.12   11.88   library setup time
          11.88   data required time
---------------------------------------------------------
          11.88   data required time
          -3.32   data arrival time
---------------------------------------------------------
           8.56   slack (MET)



========== WNS ==========
worst slack max -17.16

========== TNS ==========
tns max -5357.16
```

4. Summary of All 16 Corners

#	Corner	Temperature	Voltage	Arrival (ns)	Slack (ns)
1	FF	100°C	1.65 V	2.64	9.26
2	FF	100°C	1.95 V	2.05	9.88
3	FF	−40°C	1.56 V	3.46	8.40
4	FF	−40°C	1.65 V	2.98	8.90
5	FF	−40°C	1.76 V	2.56	9.34
6	FF	−40°C	1.95 V	2.08	9.84
7	SS	100°C	1.40 V	9.93	1.57
8	SS	100°C	1.60 V	6.76	4.95
9	SS	−40°C	1.28 V	27.48	−17.16
10	SS	−40°C	1.35 V	18.81	−7.89
11	SS	−40°C	1.40 V	15.11	−3.95
12	SS	−40°C	1.44 V	12.95	−1.65
13	SS	−40°C	1.60 V	8.08	3.53
14	SS	−40°C	1.76 V	5.78	5.97
15	TT	25°C	1.80 V	3.41	8.46
16	TT	100°C	1.80 V	3.32	8.56

Reported paths: 12 MET and 4 VIOLATED. All four violations occur at SS, −40°C, from 1.28 V through 1.44 V. These are path-specific results, not proof that every other path passes.

5. WNS, TNS and Interpretation

Global WNS (worst negative slack): −17.16 ns. Global TNS (sum of negative endpoint slacks): −5357.16 ns. These values repeat for all corners because the report_worst_slack -max and report_tns -max commands were not corner-selected. Do not label them as separate WNS/TNS values for each corner.

The SS −40°C, 1.28 V path is the most severe reported setup violation (arrival 27.48 ns; required 10.32 ns). FF and TT reported paths meet timing. Increasing voltage markedly improves the SS results.

6. Conclusion and Limitations

Multi-corner post-synthesis STA of VSDBabySoC using 16 Sky130HD PVT libraries identified four low-voltage SS setup violations for a 12 ns clock. The worst reported slack is −17.16 ns, confirming that the design does not meet this timing target across all analyzed corners. This analysis uses ideal clock delays and no extracted routing parasitics; post-route STA, hold checks, and validation of the SDC/PLL clock constraints are still needed for signoff.